Algebra Monsoon 2026 Lecture 14 Canonical isomorphisms

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Canonical isomorphisms.

The canonical decomposition of a morphism $f: G \to H$ of groups:

Let $f_*(G) \subset H$ be the image of f and $f^*(\{e\}) \subset G$ the kernel of f.

We have the inclusion $i: f_*(G) \to H$ and the quotient $\pi: G \to G/f^*(\{e\})$.

By the universal property of the quotient, there is a unique morphism $\overline{f}: G/f^*(\{e\}) \to f_*(G)$ such that $f = \overline{f} \circ \pi$.

- $\overline{f}$ is an isomorphism.

Proof: It is obviously surjective. If $\overline{f}(a) = \overline{f}(b)$ for some $a, b \in G/f^*(\{e\})$, writing $a = \pi(x)$ and $b = \pi(y)$ for some $x, y \in G$, we would have $f(x) = f(y)$, so $f(xy^{-1}) = e$, so $xy^{-1} \in f^*(\{e\})$, so $\pi(xy^{-1}) = e$, so $\pi(x) = \pi(y)$, so $a = b$. Bijective morphisms of groups are isos, □

so we are done.

The decomposition $f = i \circ \overline{f} \circ \pi$ is called the <u>canonical decomposition</u> of the morphism of groups f.

Let G be a group, $N \subset G$ a normal subgroup, and $\pi : G \to G/N$ the quotient.

<u>Proposition</u> If a subgroup $H \subset G$ contains N, then N is normal in H and there is a unique isomorphism $\iota : H/N \xrightarrow{\sim} \pi_*(H)$ making the triangle

$$\begin{array}{ccc} H & \longrightarrow & H/N \\ & \searrow^{\pi|_H} & \downarrow \iota \\ & & \pi_*(H) \end{array}$$

commute.

<u>Proof</u> As $\pi|_H$ is a morphism of groups whose kernel is $\ker(\pi) \cap H = N$, there is a unique morphism of groups $\iota : H/N \to \pi_*(H)$ making the triangle commute. This ι is injective because $\ker(\pi|_H) = N$, and it is surjective because $\pi|_H$ is surjective. □

- The mapping $C \longmapsto \pi_*(C)$ is a bijection

(subgroups of G containing N)

$\downarrow$

(subgroups of G/N);

the reciprocal bijection is $D \mapsto \pi^*(D)$.

Proof Clearly $\pi_*(C)$ is a subgroup of G/N for every subgroup C of G, and $\pi^*(D)$ is a subgroup of G for every subgroup D of G/N. We have $\pi_*(\pi^*(D)) = D$ because π is surjective. For every subgroup C of G containing N, $\pi^*(\pi_*(C))$ is a subgroup of G containing C; let us show equality. Let $a \in \pi^*(\pi_*(C))$. Then $\pi(a) \in \pi_*(C)$, so there is a $c \in C$ such that $\pi(a) = \pi(c)$. Put $b = ac^{-1}$ we have $\pi(b) = e$, so $b \in N$, so $b \in C$, so $a = bc$ is in C. Therefore $C = \pi^*(\pi_*(C))$.

□

- A subgroup C of G containg N is normal in G if and only if $\pi_*(C)$ is normal in G/N. When this is so, the unique morphism $\iota : G/C \to (G/N)/\pi_*(C)$ making the square

$$\begin{array}{ccc} G & \longrightarrow & G/C \\ {\scriptstyle \pi}\downarrow & & \downarrow{\scriptstyle \iota} \\ G/N & \longrightarrow & (G/N)/\pi_*(C) \end{array}$$

Commute is an isomorphism.

Proof Suppose that C is normal in G, and let $x \in G/N$, $y \in \pi_*(C)$. Write $x = \pi(a)$ and $y = \pi(b)$ for some $a \in G$, $b \in C$. Then $aba^{-1} \in C$, hence $xyx^{-1} = \pi(aba^{-1})$ is in $\pi_*(C)$.

Conversely, if $\pi_*(C)$ is normal in G/N, then the kernel of the composite morphism

$$G \longrightarrow G/N \longrightarrow (G/N)/\pi_*(C)$$

is C, therefore C is normal in G and ι is injective

As this composite is surjective, so is ι. □

Let G be a group and $N \subset G$ a normal subgp.
For every subgroup $H \subset G$, we have

- $HN = NH$, and it is a subgroup of G
- H is a normal subgroup of HN.
- $H \cap N$ is a normal subgroup of H.
- The unique morphism $\iota : H/(H \cap N) \to HN/N$

making the square

$$\begin{array}{ccc} N & \longrightarrow & N/(N \cap H) \\ \downarrow & & \downarrow \iota \\ NH & \longrightarrow & NH/N \end{array}$$

commute is an isomorphism.

Proof Let $\pi : G \to G/N$ be the quotient of G by N. Then the kernel of $\pi|_H$ is $H \cap N$, so $H \cap N$ is normal in H, and there is an induced isomorphism $H/(H \cap N) \xrightarrow{\sim} \pi_*(H)$.

Next, we obviously have

$$NH \subset \pi^*(\pi_*(H));$$

to show equality, let $x \in \pi^*(\pi_*(H))$.

There exists $h \in H$ such that $\pi(x) = \pi(h)$.

Put $a = xh^{-1}$. Then $\pi(a) = e$, so

$a \in N$, and hence $x = ah$ is in NH.

In particular, NH is a subgroup containing N, and $\pi|_{NH}$ induces an isomorphism $NH/N \xrightarrow{\sim} \pi_*(H)$.

The morphism c making the square commute is the composite

$$H/(H \cap N) \xrightarrow{\sim} \pi_*(H) \xrightarrow{\sim} NH/N,$$

hence it is an isomorphism. $\square$